

Future Models Roadmap and Final Project Recommendations

This document outlines the remaining models (6–9) to be implemented in the Amazon Fine Food Reviews sentiment analysis project, including academic justification, expected contributions, business relevance, and final recommendations.

6. AdaBoost

Why This Model?

AdaBoost is an ensemble learning technique that improves weak learners by focusing on previously misclassified observations.

Expected Contribution

Improve classification performance by correcting errors made in earlier iterations.

Business Relevance

Useful for identifying difficult customer reviews that simpler models may misclassify.

Strengths

Simple boosting approach, interpretable, efficient.

Weaknesses

Sensitive to noisy data and outliers.

Professor Defense

AdaBoost was selected to evaluate whether sequential boosting can outperform bagging-based methods such as Random Forest.

7. XGBoost / LightGBM

Why This Model?

These are advanced gradient boosting algorithms widely used in industry and machine learning competitions.

Expected Contribution

Capture complex interactions among TF-IDF features and improve predictive performance.

Business Relevance

Supports robust and scalable sentiment classification for large datasets.

Strengths

High accuracy, feature importance, scalability.

Weaknesses

Less interpretable than simpler models.

Professor Defense

Chosen because gradient boosting methods consistently achieve top performance on structured machine learning tasks.

8. LSTM

Why This Model?

LSTM is designed for sequential data and can understand word order and context.

Expected Contribution

Capture linguistic patterns that traditional machine learning models may miss.

Business Relevance

Provides deeper understanding of customer opinions expressed in reviews.

Strengths

Handles sequential information and contextual dependencies.

Weaknesses

Requires greater computational resources and training time.

Professor Defense

Included to evaluate the benefits of deep learning for sentiment classification.

9. BERT / DistilBERT

Why This Model?

Transformer-based models represent the current state of the art in NLP.

Expected Contribution

Deliver the highest contextual understanding and sentiment classification performance.

Business Relevance

Enables highly accurate customer sentiment analysis for decision-making.

Strengths

Superior contextual understanding, strong generalization.

Weaknesses

Computationally expensive compared with traditional models.

Professor Defense

Selected to represent modern transformer-based NLP and establish the project's performance ceiling.

Expected Project Evolution

The project progresses through linear models, probabilistic models, margin-based models, tree-based models, ensemble methods, deep learning architectures, and transformer-based models. This allows a comprehensive comparison between interpretability, computational complexity, and predictive performance.

Final Recommendation

The final objective is not only to identify the highest-performing model but also to understand the trade-offs between model complexity, interpretability, computational cost, and business value. Traditional machine learning models provide strong baselines, ensemble methods improve robustness, deep learning captures sequential patterns, and transformer models deliver state-of-the-art language understanding.